Polynomial Regression — Assignment 1 Report

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Problem	Model submitted	Terms used	CV MSE (± s.e.)	CV R^2
var1 — Net Power Score (x1…x6)	Total degree **5**, Lasso α=0.005	158 of 462	0.358 ± 0.009	0.9662
var2 — Thermal Anomaly Score (x1…x3)	Total degree **8**, Lasso α=0.001	155 of 165	0.271 ± 0.006	0.9939

CV = 5-fold cross-validation repeated 3 times on the 1000 training rows (15 fits per configuration). R^2 is 1 – CV MSE / Var(y).

1. Data and what it implies for modelling

Each problem has 1000 training and 1000 test rows, with no missing values or duplicates. All inputs lie in [–1, 1]. Var(y) is 10.60 for var1 and 44.64 for var2. Two observations shaped the method:

- **Inputs are clipped to ±1.** 31% of var1 and 24% of var2 training values are exactly ±1. Out-of-fold residuals of a good var1 model are about the same whether 0, 1, 2 or 3 coordinates are clipped (MSE ≈ 0.34–0.37). This indicates y was generated from the clipped values, so no special handling is needed.
- **The var1 test set is shifted toward the corners of the cube.** 50% of var1 test values are ±1 (vs 31% in training). 65% of test rows have ≥3 clipped coordinates, against 28% of training rows (Fig. 1). The edges of $[–1, 1]^6$ are where high-degree polynomials oscillate most, so validation error is also reported *re-weighted to the test distribution*. var2 shows almost no such shift.

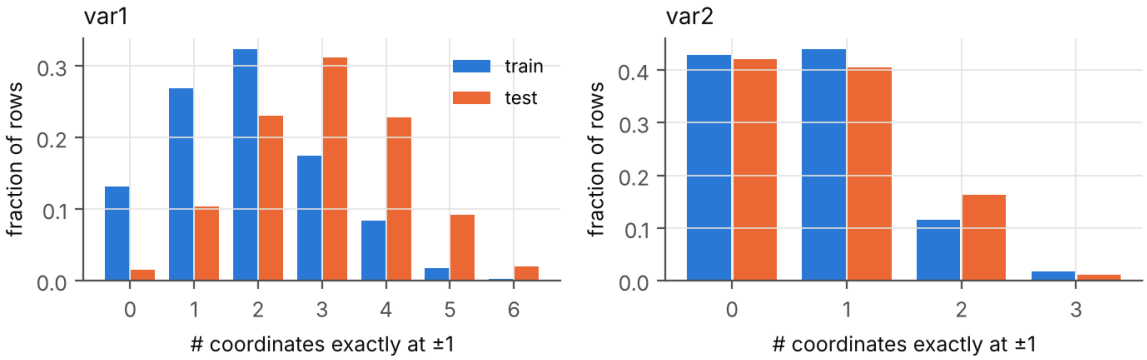


Fig. 1 — Share of rows by number of coordinates exactly at ±1. var1's test set is concentrated near the corners; var2's matches its training set.

2. Method

2.1 Polynomial features in a well-conditioned basis

A polynomial of total degree d contains every term $x1^a x2^b \ldots$ with a+b+… ≤ d. For k inputs that is C(k+d, d) terms: 462 for var1 at d=5, 165 for var2 at d=8, and 1771 for var2 at d=20. Raw monomials become nearly collinear at high degree (x^{18} and x^{20} look almost the same on [–1, 1]). So I build the same space from products of Legendre polynomials $P_a(x1)\cdot P_b(x2)\cdot\ldots$ with a+b+… ≤ d. Each P_a is a degree-a polynomial and the set is a basis of the same space. The fitted function is therefore still an ordinary polynomial of degree d, and `export_equation.py` converts it back to monomial coefficients. Only the numerics change: on the training data the design-matrix condition number drops from 433 to 181 (var1, d=5), from 1.2e+03 to 24 (var2, d=8) and from 9.3e+19 to 9.7e+16 (var2, d=20). This makes the high-degree end of the search reliable.

2.2 Estimators

For each degree, four ways of fitting the coefficients are compared. All of them produce a polynomial of that degree. **OLS** is plain least squares. **Ridge** adds an L2 penalty α■β■² (intercept not penalised). **Lasso** adds an L1 penalty α■β■₁, which sets many coefficients exactly to zero and so selects which terms of the degree-d polynomial to keep. **Relaxed Lasso** uses the Lasso only to choose terms, then refits them by OLS to remove the shrinkage bias. Penalties are what make degrees with as many terms as rows fittable at all, so they decide how far up the degree ladder the search can go.

2.3 Model selection

- **Search space:** var1 degrees 1–8 (degree 8 = 3003 terms, already 3× the sample size); var2 degrees 1–20, the full range the problem statement allows. Ridge uses 6–7 values of α at every degree. Lasso and relaxed Lasso use 4–6 values of α up to degree 7 (var1) or 12 (var2); above that the curves were already rising and the cost was high. OLS is skipped once the number of terms exceeds the fold's training rows.

- **Score:** mean MSE over 5-fold × 3-repeat CV, with the same fixed splits for every configuration. There is also a test-weighted variant: each validation error is multiplied by $w(k) = P_{test}(k)/P_{train}(k)$, where k is the number of clipped coordinates (capped at 10).
- **Rule:** take the configuration with the lowest test-weighted CV MSE. Among all configurations within one standard error of it, choose the *lowest degree*, then the best score at that degree. The final model is refit on all 1000 training rows.

3. Problem var1 — six turbine settings

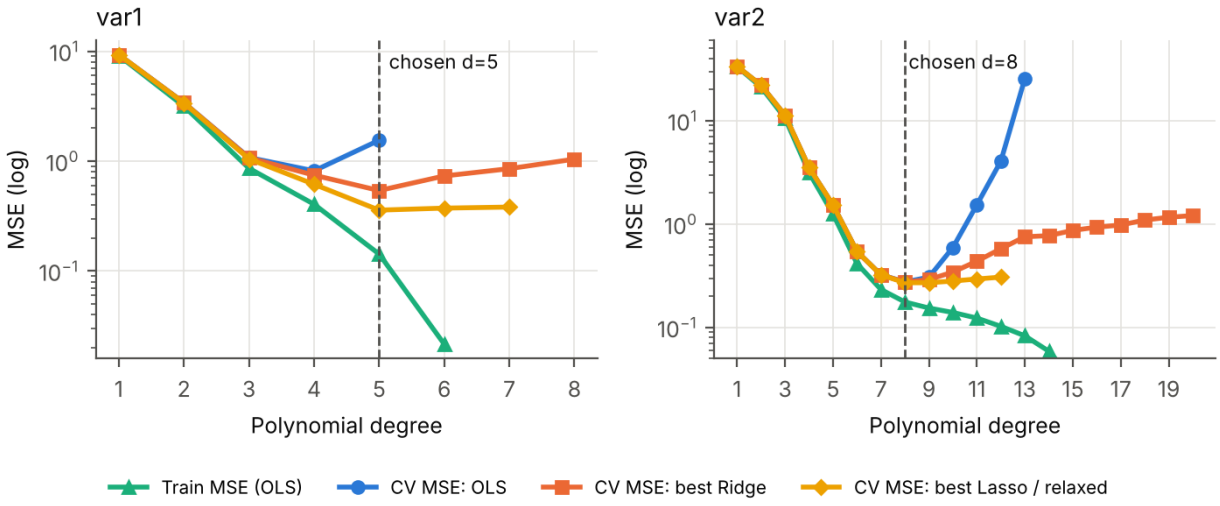


Fig. 2 — Training and CV error against degree (log scale). Training error keeps falling; CV error has a clear minimum. Plain OLS overfits one or two degrees earlier than penalised fits. var2's Lasso curve stops at d=12, where the search for it ended.

Deg.	# terms	OLS CV MSE	Best estimator	Best CV MSE	Test-weighted
1	7	9.293	Lasso (α=0.02)	9.288	14.394
2	28	3.398	Lasso (α=0.01)	3.377	6.126
3	84	1.078	Lasso (α=0.005)	1.046	1.355
4	210	0.815	Relaxed Lasso (α=0.02)	0.615	0.795
5	462	1.548	Lasso (α=0.005)	0.358	0.387
6	924	n/a (p ≥ n)	Lasso (α=0.005)	0.373	0.410
7	1716	n/a (p ≥ n)	Lasso (α=0.005)	0.382	0.414
8	3003	n/a (p ≥ n)	Ridge (α=10)	1.042	1.168

Table 1 — var1: best configuration at each degree (CV MSE). Highlighted row = submitted model.

Plain OLS is best at degree 4 (CV MSE 0.815). At degree 5 OLS has 462 coefficients for 800 training rows per fold and its error rises to 1.55. Fig. 3 shows that the L1 penalty recovers degree 5 and more than halves the error, while the L2 penalty helps much less. Ridge keeps all 462 terms and only shrinks them. The Lasso keeps about a third of the terms, which suggests the true degree-5 polynomial is sparse. Degrees 6 and 7 with Lasso are no better (0.373, 0.382): the extra terms are not needed. Degree 3 or 4 is clearly too simple (1.05, 0.61). So the submitted model is **degree 5, Lasso α=0.005, with 158 non-zero terms**.

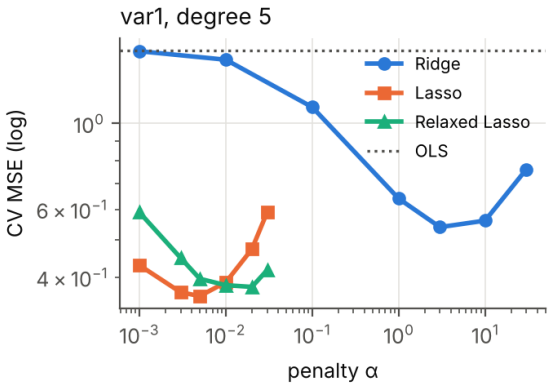


Fig. 3 — var1 at degree 5: CV MSE against penalty strength for each estimator.

Are all six inputs needed? Removing one input at a time and refitting the chosen configuration gives CV MSE of x1: 2.88, x2: 2.56, x3: 3.98, x4: 1.54, x5: 3.91, x6: 4.73 (vs 0.36 with all six). Every input is needed: dropping any one multiplies the error by 4–13×. The model therefore uses all six settings.

Noise floor. Training MSE is 0.24 and CV MSE is 0.36. Out-of-fold error is flat across clipping groups and across degrees 5–7, so about 0.35 appears to be irreducible noise rather than missing structure.

4. Problem var2 — 3-D thermal anomaly field

var2 has only three inputs, so a high degree is affordable (165 terms at d=8, 1771 at d=20). The right panel of Fig. 2 shows a textbook bias–variance curve. CV error drops by two orders of magnitude up to d≈8, and OLS then explodes (d=13: MSE 26; d≥15: no unique solution). Penalties keep the high-degree fits under control but never beat degree 8. At d=20 the best Ridge fit has 4–5× the error of d=8, so the full degree budget is clearly not needed.

Deg.	# terms	OLS CV MSE	Best estimator	Best CV MSE	Test-weighted
4	35	3.527	Lasso (α=0.001)	3.526	3.593
5	56	1.529	Ridge (α=0.1)	1.529	1.544
6	84	0.544	Relaxed Lasso (α=0.01)	0.543	0.563
7	120	0.323	Lasso (α=0.0003)	0.322	0.320
8	165	0.276	Lasso (α=0.001)	0.271	0.272
9	220	0.306	Lasso (α=0.001)	0.271	0.269
10	286	0.593	Lasso (α=0.001)	0.281	0.281
12	455	4.057	Lasso (α=0.001)	0.307	0.307
14	680	n/a (p ≥ n)	Ridge (α=1)	0.772	0.811
16	969	n/a (p ≥ n)	Ridge (α=1)	0.933	0.987
20	1771	n/a (p ≥ n)	Ridge (α=1)	1.208	1.276

Table 2 — var2: best configuration at selected degrees (CV MSE). Highlighted row = submitted model.

Degrees 8 and 9 tie (difference 0.0001, under 1/80 of one standard error), and degree 7 is 0.05 worse. Under the one-standard-error rule **degree 8** is chosen. Unlike var1, the var2 polynomial is *dense*: the Lasso keeps 155 of 165 terms, and OLS, Ridge and Lasso at d=8 are within 0.005 of each other. A light Lasso (α=0.001) has a marginally lower test-weighted error, so it is used. All three inputs matter: removing one gives CV MSE x1: 24.2, x2: 38.2, x3: 17.4 (vs 0.27).

5. Final checks

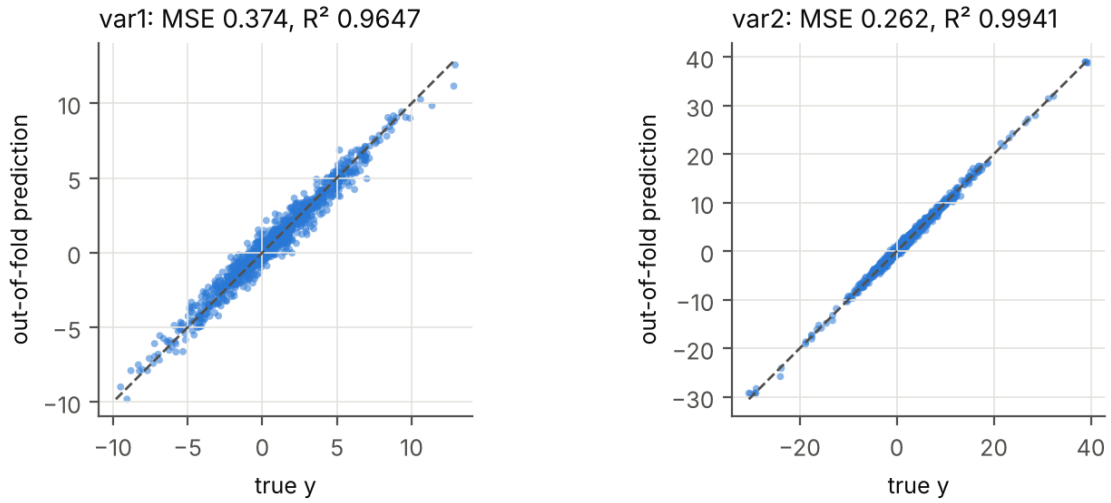


Fig. 4 — Out-of-fold predictions of the submitted models against the true y (training set).

- **Residuals** scatter evenly around the diagonal across the whole range of y, including the extremes (Fig. 4). There is no curvature, which would have been a sign of underfitting.
- **Robustness on the test inputs.** RMS difference between the submitted predictions and those of the runner-up models: var1: d=5 relaxed: 0.27, d=6 lasso: 0.12; var2: d=9 lasso: 0.15, d=8 ols: 0.08. All are below the noise standard deviation (≈0.5–0.6), so the predictions do not depend on the exact pick. For var1 I also checked that predictions in each clipping group (k = 0…6) stay within the training y range of that group. They do, so the model does not blow up at the corners even though the test set is corner-heavy.

- **Generalisation estimate.** Expected test MSE ≈ 0.39 (var1, test-weighted) and 0.27 (var2). Expected test R^2 ≈ 0.963 and 0.994.

6. Reproducibility

Repository: https://github.com/Mitansh1309/Polynomial-Regression. `bash run_all.sh` runs the full grid search (≈5 min), refits the final models, writes `predictions/BT2024046_pred_var1.csv` and `BT2024046_pred_var2.csv`, and regenerates every figure. All CV results are saved in `results/cv_var*.csv`. All randomness uses fixed seeds. Dependencies: numpy, pandas, scikit-learn (Lasso solver only), matplotlib.